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IMMUNOBIOLOGY AND IMMUNOTHERAPY

Comment on Thwaites et al, page 820

RAGs usurp cellular factors for both breaking and repairing

Katheryn Meek | Michigan State University

There is clear evidence that the recombination-activating gene (RAG) complex functions best in vitro with help from the high-mobility group box 1 (HMGB1) DNA-bending protein. In this issue of *Blood*, Thwaites et al show unequivocally that HMGB1 is similarly required for RAG cleavage in vivo.[1]

The unparalleled ingenuity of the V(D)J recombination reaction (that allows millions of developing lymphocytes to express their own unique immune receptor on the cell surface) has captivated scientists for decades.[2] The mechanism that provides for this vast repertoire of unique antigen receptors is a genetic recombination system that functions only in developing lymphocytes to assemble exons that encode the antigen binding domain to assemble antigen binding exons from a small repertoire of gene segments. Recombination requires a lymphocyte-specific endonuclease (the products of RAG1 and RAG2). V(D)J recombination is initiated by RAG binding recombination signal sequences (12-RSS and 23-RSS) to generate a paired 12-/23-RSS complex (12/23 rule). Cleavage occurs in 2 steps: first, a single-strand nick at the heptamer coding end juncture, and then a transesterification reaction to generate blunt signal ends and covalently sealed hairpinned coding ends. Emerging structural studies reveal how the RAGs coordinate coupled cleavage of paired 12-RSS and 23-RSS.[3,4] Although the RAG endonuclease is the only requisite lymphocyte-specific component of the V(D)J complex, the RAG complex provides only a part of the biochemical basis for V(D)J recombination. It is well understood that the RAG endonuclease recruits factors from the classical nonhomologous end joining (c-NHEJ) pathway to repair its DNA double-strand breaks (DSBs). Joining these breaks using this error-prone pathway ensures that every time a particular set of gene segments is recombined, additional random sequence diversity occurs at the junctions, which (in the assembled immune receptor gene) encode the actual antigen binding pocket, vastly enhancing antigen binding diversity. Moreover, it is clear (but not understood by what mechanism) that the RAG complex functions after cleavage to restrict access to its DSBs so that only the c-NHEJ pathway can repair the DSBs.[2,5]

Thwaites et al have now provided good data to support the hypothesis that the RAGs commandeer another ubiquitous factor (HMGB1) to facilitate DNA cleavage. HMGB1 is an abundant DNA chromatin factor that can facilitate many cellular functions by bending DNA and promoting assembly of nucleoprotein complexes. Early seminal studies[6] demonstrated that HMGB1 dramatically enhances RAG cleavage, presumably by its ability to induce bending of the RSSs, especially the 23-RSS. Although RAGs (in the presence of manganese) efficiently nick and cleave isolated 12-RSS or 23-RSS,[7] coupled cleavage of paired 12-RSS and 23-RSS in the presence of the physiologic cation magnesium is dramatically more efficient in the presence of HMGB1. These in vitro data suggest that HMGB1 might be an essential cofactor for RAG cleavage in vivo, analogous to how DNA-bending proteins facilitate site-specific recombination in prokaryotes.[6] This hypothesis is diminished by studies that document normal lymphocyte development in HMGB1-deficient mice.[8] The possibility of functional redundancy between HMGB1 and HMGB2, which are 79% identical and are both expressed in the developing lymphocytes assessed in this study, has left the requirement for HMGB1 in V(D)J recombination an open controversy. The report by Thwaites et al may just end this controversy.

Thwaites et al carefully analyzed the compound RAG1 mutations identified in a patient with primary immunodeficiency. A mutation on 1 RAG allele is hypomorphic (R504Q), which explains the relatively mild immunodeficiency observed in the patient. The mutation on the other allele, R401W, dramatically impairs RAG function.

In a series of in vitro and cellular experiments, the authors demonstrate that R401W specifically disrupts RAG binding to HMGB1. HMGB1 binds the RSS spacer in RAG-DNA complexes. HMGB1 contains 2 DNA binding sites, Box A and Box B. From structural studies of synaptic RAG tetramers, including HMGB1,[3,9] binding of the 12-RSS requires only 1 binding site (Box A), whereas binding of the 23-RSS requires both binding sites. Moreover, Box A of HMGB1 seems to be directly opposed to R401W at both 12-RSS and 23-RSS in the synaptic complex. The study by Thwaites et al demonstrates that the R401W mutation disrupts HMGB1 binding to RAG tetramers, cripples RAG-coupled cleavage in vitro, and virtually ablates recombination in cellular V(D)J assays. Remarkably, the R401W

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mutant performs uncoupled cleavage in a way similar to that of wild-type RAG1. These findings provide strong support for a stringent requirement for HMGB1 during V(D)J recombination. In complementary approaches, HMGB1 was partially silenced (in a cell strain that lacks HMGB2), resulting in severe reduction of V(D)J recombination in cellular assays. Moreover, HMGB1 or HMGB2 complementation restored V(D)J recombination in a dose-dependent manner with wild-type RAGs but not with the R401W mutant.

The R401W mutation clearly ablates coupled cleavage while leaving uncoupled cleavage intact in vitro. It may be useful to study the R401W mutation in cellular or mouse models of chromosomal V(D)J recombination to determine whether uncoupled cleavage can occur in vivo (potentially not feasible because of the requirement for manganese). If uncoupled cleavage occurs with this mutant in vivo, it will be of considerable interest to determine whether the single DSBs introduced are also tightly restricted to c-NHEJ. It seems possible that structural reorganization of the RAG complex induced by HMBG1 binding might have an impact on how the RAG complex cooperates with factors of both the c-NHEJ pathway and the DNA damage response pathway to prevent genomic instability that can result if V(D)J recombination is dysregulated.

Thus, just as its ancestral transposons (regarded by many as DNA parasites) co-opt cellular factors to spread and propagate, the RAG complex hijacks factors that facilitate both the cleavage and joining facets of V(D)J recombination.

Conflict-of-interest disclosure: The author declares no competing financial interests. ■

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LYMPHOID NEOPLASIA

Comment on Close et al, page 830

FBXW7 is a biologically validated cancer driver gene for CLL

Davide Rossi | Oncology Institute of Southern Switzerland

F-box and WD40 repeat domain containing-7 (*FBXW7*) is one of the statistically defined candidate cancer driver genes for chronic lymphocytic leukemia (CLL). In this issue of *Blood*, Close et al describe functional genomic experiments that have verified the biological consequences of mutations in *FBXW7*, which can now be considered a validated driver gene for CLL.[1]

The mere presence of somatic mutations is insufficient to implicate a gene in cancer. For example, when analytical methods are applied to whole-exome sequencing data from a given cancer, almost a quarter of the somatically mutated genes encode proteins that are irrelevant for tumor initiation or progression, such as olfactory receptors, cardiac ryanodine receptors (RyR2, RyR3), cytoskeletal dyneins (DNAH5, DNAH11), or neuronal synaptic vesicle proteins (PCLO).[2]

Cancer geneticists and bioinformaticians use the term "passenger" to describe such mutations that were likely being randomly acquired to distinguish them from mutations targeting candidate cancer driver genes that were implicated in the tumor biology. In the absence of biological data, a statistical definition of candidate cancer driver genes was created and then used by cancer geneticists and bioinformaticians. Any given gene was labeled as a candidate cancer driver if it harbored somatic point mutations (that is, substitutions and small insertion or deletions) at a statistically significant rate or pattern in cancer tumor samples. Three independent statistical signals are used to separate candidate cancer driver genes from other genes: (1) they have a high mutational burden relative to background expectation, (2) there is a clustering of mutations within the gene, and (3) mutations are enriched in the evolutionarily conserved amino acids of the corresponding proteins.[3]

In CLL, more than 40 genes fulfill the statistical definition of candidate cancer driver, but only a few of them are biologically validated as oncogenes or tumor-suppressor genes (ie, *SF3B1*, *NOTCH1*, *TP53*, *ATM*).[4]

FBXW7 encodes an E3 ubiquitin ligase whose WD40 domain interacts with proteins that are subsequently subject to proteasomal degradation (see figure panel A). Among FBXW7 targets for proteasomal degradation, there are a number of known proto-oncogenes, such as cleaved NOTCH1 (NOTCH1-NOTCH1 intracellular domain [NICD]), MYC, and NF-κB2.[5] In CLL, *FBXW7* mutations occur at a frequency of 2% to 6% and commonly affect the WD40 substrate binding domain, including hotspot residues that are required for substrate recognition. Close et al identify cleaved

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